

Chapter 3 Tokens Assignment

1. C99 standard guarantees uniqueness of _____ characters for internal names.
 - a) 31
 - b) 63
 - c) 12
 - d) 14
2. C99 standard guarantees uniqueness of _____ characters for external names.
 - a) 31
 - b) 6
 - c) 12
 - d) 14
3. Which of the following is not a valid variable name declaration?
 - a) int __a3;
 - b) int __3a;
 - c) int __A3;
 - d) None of the mentioned
4. Which of the following is not a valid variable name declaration?
 - a) int _a3;
 - b) int a_3;
 - c) int 3_a;
 - d) int _3a
5. Variable names beginning with underscore is not encouraged. Why?
 - a) It is not standardized
 - b) To avoid conflicts since assemblers and loaders use such names
 - c) To avoid conflicts since library routines use such names
 - d) To avoid conflicts with environment variables of an operating system
6. All keywords in C are in
 - a) LowerCase letters
 - b) UpperCase letters
 - c) CamelCase letters
 - d) None
7. Variable name resolving (number of significant characters for uniqueness of variable) depends on
 - a) Compiler and linker implementations
 - b) Assemblers and loaders implementations
 - c) C language
 - d) None
8. Which of the following is not a valid C variable name?
 - a) int number;
 - b) float rate;
 - c) int variable_count;
 - d) int \$main;
9. Which of the following is true for variable names in C?
 - a) They can contain alphanumeric characters as well as special characters
 - b) It is not an error to declare a variable to be one of the keywords (like goto, static)
 - c) Variable names cannot start with a digit
 - d) Variable can be of any length
10. Which is valid C expression?
 - a) int my_num = 100,000;
 - b) int my_num = 100000;
 - c) int my num = 1000;
 - d) int \$my_num = 10000;
11. Which of the following is not a valid variable name declaration?
 - a) float PI = 3.14;
 - b) double PI = 3.14;
 - c) int PI = 3.14;
 - d) #define PI 3.14
12. What will happen if the below program is executed?

```
#include <stdio.h>
main()
{
    int main = 3;
    printf("%d", main);
}
```

 - a) It will cause a compile-time error
 - b) It will cause a run-time error
 - c) It will run without any error and prints 3
 - d) It will experience infinite looping

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13. Which of the following cannot be a variable name in C?

- a) volatile
- b) true
- c) friend
- d) export

14. The format identifier '%i' is also used for _____ data type?

- a) char
- b) int
- c) float
- d) double

15. Which data type is most suitable for storing a number 65000 in a 32-bit system?

- a) signed short
- b) unsigned short
- c) long
- d) int

16. What is the size of an int data type?

- a) 4 Bytes
- b) 8 Bytes
- c) Depends on the system/compiler
- d) Cannot be determined

17. What is the output of this C code?

```
signed char chr;  
chr = 128;  
printf("%d\n", chr);
```

- a) 128
- b) -128
- c) Depends on the compiler
- d) None of the mentioned

18. What is short int in C programming?

- a) Basic datatype of C
- b) Qualifier
- c) short is the qualifier and int is the basic datatype
- d) All of the mentioned

19. What is the output of this C code?

```
int a = 1;  
if (a--)
```

```
printf("True");  
if (a++)  
printf("False");
```

- a) True
- b) False
- c) True False
- d) No Output

20. Comment on the output of this C code?

```
float f1 = 0.1;  
if (f1 == 0.1)  
printf("equal\n");  
else  
printf("not equal\n");
```

- a) equal
- b) not equal
- c) Output depends on compiler
- d) None of the mentioned

21. Comment on the output of this C code?

```
float f1 = 0.1f;  
if (f1 == 0.1f)  
printf("equal\n");  
else  
printf("not equal\n");
```

- a) equal
- b) not equal
- c) Output depends on compiler
- d) None of the mentioned

22. Which is correct with respect to size of the datatypes?

- a) char > int > float
- b) int > char > float
- c) char < int < double
- d) double > char > int

23. What is the output of this C code?

```
float x = 'a';  
printf("%f", x);
```

- a) a
- b) run time error
- c) a.0000000
- d) 97.000000

24. What is the output of this C code?

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```
enum {ORANGE = 5, MANGO,
BANANA = 4, PEACH};
printf("PEACH = %d\n", PEACH);
```

- a) PEACH = 3
- b) PEACH = 4
- c) PEACH = 5
- d) PEACH = 6

25. What is the output of this C code?

```
printf("C programming %s", "Class
by\n%s Sanfoundry", "WOW");
```

- a) C programming Class by
WOW Sanfoundry
- b) C programming Class by\n%s
Sanfoundry
- c) C programming Class by
%s Sanfoundry
- d) Compilation error

26. What is the output of this C code?

```
#define a 10
int main()
{
    const int a = 5;
    printf("a = %d\n", a);
}
```

- a) a = 5
- b) a = 10
- c) Compilation error
- d) Runtime error

27. What is the output of this C code?

```
int var = 010;
printf("%d", var);
```

- a) 2
- b) 8
- c) 9
- d) 10

28. What is the output of this C code?

```
enum birds {SPARROW, PEACOCK,
PARROT};
enum animals {TIGER = 8, LION,
RABBIT, ZEBRA};
main()
```

```
{
    enum birds m = TIGER;
    int k;
    k = m;
    printf("%d\n", k);
}
```

- a) 0
- b) Compile time error
- c) 1
- d) 8

29. What is the output of this C code?

```
#define MAX 2
enum bird {SPARROW = MAX + 1,
PARROT = SPARROW + MAX};
main()
{
    enum bird b = PARROT;
    printf("%d\n", b);
}
```

- a) Compilation error
- b) 5
- c) Undefined value
- d) 2

30. enum types are processed by

- a) Compiler
- b) Preprocessor
- c) Linker
- d) Assembler

31. What is the output of this C code?

```
printf("sanfoundry\rclass\n");
```

- a) sanfoundryclass
- b) sanfoundry
class
- c) classundry
- d) sanfoundry

32. What is the output of this C code?

```
printf("sanfoundry\r\n\rclass\n");
```

- a) sanfoundryclass
- b) sanfoundry
class
- c) classundry

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d) sanfoundry

33. What is the output of this C code?

```
const int p;  
p = 4;  
printf("p is %d", p);
```

- a) p is 4
- b) Compile time error
- c) Run time error
- d) p is followed by a garbage value

34. Which keyword is used to prevent any changes in the variable within a C program?

- a) immutable
- b) mutable
- c) const
- d) volatile

35. What is the output of this C code?

```
int i = 3;  
int l = i / -2;  
int k = i % -2;  
printf("%d %d\n", l, k);
```

- a) Compile time error
- b) -1 1
- c) 1 -1
- d) Implementation defined

36. Which of the following data type will throw an error on modulus operation(%)?

- a) char
- b) short
- c) int
- d) float

37. What is the output of this C code?

```
int a = 10;  
double b = 5.6;  
int c;  
c = a + b;  
printf("%d", c);
```

- a) 15
- b) 16
- c) 15.6

d) 10

38. What is the output of this C code?

```
int a = 10, b = 5, c = 5;  
int d;  
d = a == (b + c);  
printf("%d", d);
```

- a) Syntax error
- b) 1
- c) 10
- d) 5

39. What is the output of this C code?

```
int x = 1, y = 0, z = 5;  
int a = x && y || z++;  
printf("%d", z);
```

- a) 6
- b) 5
- c) 0
- d) Varies

40. What is the output of this C code?

```
int x = 1, y = 0, z = 5;  
int a = x && y && z++;  
printf("%d", z);
```

- a) 6
- b) 5
- c) 0
- d) Varies

41. What is the output of this C code?

```
int x = 0, y = 2, z = 3;  
int a = x & y | z;  
printf("%d", a);
```

- a) 3
- b) 0
- c) 2
- d) Run time error

42. What is the final value of j in the below code?

```
int i = 0, j = 0;  
if (i && (j = i + 10))  
    //do something  
    ;
```

- a) 0
- b) 10
- c) Depends on the compiler

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d) Depends on language standard

43. What is the final value of j in the below code?

```
int i = 10, j = 0;
if (i || (j = i + 10))
    //do something
;
```

- a) 0
- b) 20
- c) Compile time error
- d) Depends on language standard

44. What is the output of this C code?

```
int i = 1;
if (i++ && (i == 1))
    printf("Yes\n");
else
    printf("No\n");
```

- a) Yes
- b) No
- c) Depends on the compiler
- d) Depends on the standard

45. What will be the value of d in the following program?

```
int a = 10, b = 5, c = 5;
int d;
d = b + c == a;
printf("%d", d);
```

- a) Syntax error
- b) 1
- c) 5
- d) 10

46. What is the output of this C code?

```
int a = 10, b = 5, c = 3;
b != !a;
c = !!a;
printf("%d\t%d", b, c);
```

- a) 5 1
- b) 0 3
- c) 5 3
- d) 1 1

47. What is the output of this C code?

```
float x = 0.1;
```

```
if (x == 0.1)
    printf("Sanfoundry");
else
    printf("Advanced C Classes");
```

- a) Advanced C Classes
- b) Sanfoundry
- c) Run time error
- d) Compile time error

48. Comment on the output of this C code?

```
float x = 0.1;
printf("%d, ", x);
printf("%f", x);
```

- a) 0.100000, junk value
- b) Junk value, 0.100000
- c) 0, 0.100000
- d) 0, 0.999999

49. What is the output of this C code?

```
int x = 97;
char y = x;
printf("%c\n", y);
```

- a) a b) b c) 97
- d) Run time error

50. What is the output of this C code?

```
unsigned int i = 23;
signed char c = -23;
if (i > c)
    printf("Yes\n");
else if (i < c)
    printf("No\n");
```

- a) Yes
- b) No
- c) Depends on the compiler
- d) Depends on the operating system

51. What is the output of this C code?

```
int i = 23;
char c = -23;
if (i < c)
    printf("Yes\n");
else
    printf("No\n");
```

- a) Yes
- b) No
- c) Depends on the compiler

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d) Depends on the standard

52. What is the output of the below code considering size of short int is 2, char is 1 and int is 4 bytes?

```
short int i = 20;
char c = 97;
printf("%d, %d, %d\n", sizeof(i),
```

```
sizeof(c), sizeof(c + i));
```

a) 2, 1, 2

b) 2, 1, 1

c) 2, 1, 4

d) 2, 2, 8

53. What will be the data type of the result of the following operation?

(float)a * (int)b / (long)c * (double)d

a) int b) long

c) float d) double

54. What is the output of this C code?

```
int a = 10, b = 10;
if (a = 5)
b--;
printf("%d, %d", a, b--);
```

a) a = 10, b = 9

b) a = 10, b = 8

c) a = 5, b = 9

d) a = 5, b = 8

55. What is the output of this C code?

```
int i = 0;
int j = i++ + i;
printf("%d\n", j);
```

a) 0

b) 1

c) 2

d) Compile time error

56. What is the output of this C code?

```
int i = 2;
int j = ++i + i;
printf("%d\n", j);
```

a) 6

b) 5

c) 4

d) Compile time error

57. What is the output of this C code?

```
int i = 0;
int x = i++, y = ++i;
printf("%d %d\n", x, y);
```

a) 0, 2

b) 0, 1

c) 1, 2

d) Undefined

58. What is the output of this C code?

```
int x = 97;
int y = sizeof(x++);
printf("X is %d", x);
```

a) X is 97

b) X is 98

c) X is 99

d) Run time error

59. What is the output of this C code?

```
int x = 4, y, z;
y = --x;
z = x--;
printf("%d%d%d", x, y, z);
```

a) 3 2 3

b) 2 3 3

c) 3 2 2

d) 2 3 4

60. What is the output of this C code?

```
int a = 5, b = -7, c = 0, d;
d = ++a && ++b || ++c;
printf("\n%d%d%d%d", a, b, c, d);
```

a) 6 -6 0 0

b) 6 -5 0 1

c) -6 -6 0 1

d) 6 -6 0 1

61. What is the output of this C code?

```
int a = -5;
int k = (a++, ++a);
printf("%d\n", k);
```

a) -4

b) -5

c) 4

d) -3

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62. What is the output of this C code?

```
int c = 2 ^ 3;
printf("%d\n", c);
```

- a) 1
- b) 8
- c) 9
- d) 0

63. What is the output of this C code?

```
unsigned int a = 10;
a = ~a;
printf("%d\n", a);
```

- a) -9
- b) -10
- c) -11
- d) 10

64. What is the output of this C code?

```
if (7 & 8)
    printf("Honesty");
if ((~7 & 0x000f) == 8)
    printf("is the best policy\n");
```

- a) Honesty is the best policy
- b) Honesty
- c) is the best policy
- d) No output

65. What is the output of this C code?

```
int a = 2;
if (a >> 1)
    printf("%d\n", a);
```

- a) 0
- b) 1
- c) 2
- d) No Output.

66. What is the output of this C code?

```
if (~0 == 1)
    printf("yes\n");
else
    printf("no\n");
```

- a) yes
- b) no
- c) Compile time error
- d) Undefined

67. What is the output of this C code?

```
int x = -2;
if (!0 == 1)
    printf("yes\n");
else
    printf("no\n");
```

- a) yes
- b) no
- c) Run time error
- d) Undefined

68. What is the output of this C code?

```
int y = 0;
if (1 |(y = 1))
    printf("y is %d\n", y);
else
    printf("%d\n", y);
```

- a) y is 1
- b) 1
- c) Run time error
- d) Undefined

69. What is the output of this C code?

```
int y = 1;
if (y & (y = 2))
    printf("true %d\n", y);
else
    printf("false %d\n", y);
```

- a) true 2
- b) false 2
- c) Either option a or option b
- d) true 1

70. What is the output of this C code?

```
int x = 0;
if (x = 0)
    printf("Its zero\n");
else
    printf("Its not zero\n");
```

- a) Its not zero
- b) Its zero
- c) Run time error
- d) None

71. What is the output of this C code?

```
int k = 8;
int x = 0 == 1 && k++;
```

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```
printf("%d%d\n", x, k);
```

- a) 0 9
- b) 0 8
- c) 1 8
- d) 1 9

72. What is the output of this C code?

```
char a = 'a';
int x = (a % 10)++;
printf("%d\n", x);
```

- a) 6
- b) Junk value
- c) Compile time error
- d) 7

73. What is the output of this C code?

```
int x = 2, y = 2;
x /= x / y;
printf("%d\n", x);
```

- a) 2
- b) 1
- c) 0.5
- d) Undefined behaviour

74. What is the output of this C code?

```
int a = -1, b = 4, c = 1, d;
d = ++a && ++b || ++c;
printf("%d, %d, %d, %d\n", a, b, c, d);
```

- a) 0, 4, 2, 1
- b) 0, 5, 2, 1
- c) -1, 4, 1, 1
- d) 0, 5, 1, 0

75. What is the output of this C code?

```
int p = 10, q = 20, r;
if (r = p = 5 || q > 20)
    printf("%d", r);
else
    printf("No Output\n");
```

- a) 1
- b) 10
- c) 20
- d) No Output

76. When 1 is entered, The output of the code below is?

```
int ch;
printf("enter a value btw 1 to 2:");
scanf("%d", &ch);
```

```
switch (ch, ch + 1)
```

```
{
    case 1:
        printf("1\n");
        break;
    case 2:
        printf("2");
        break;
}
```

- a) 1
- b) 2
- c) 3
- d) Run time error

77. What is the output of this C code?

```
int x = 0;
if (x++)
    printf("true\n");
else if (x == 1)
    printf("false\n");
```

- a) true
- b) false
- c) Compile time error
- d) Undefined behaviour